

Note 2: Workflow Triggers — Events in Detail

How Triggers Work

Every workflow file must define at least one trigger using the `on:` key at the top level. The trigger tells GitHub when to run the workflow.

Basic Workflow Structure Showing Trigger Placement

name: CI Pipeline

on:

push:

branches:

- main
- develop

jobs:

build:

runs-on: ubuntu-latest

steps:

- name: Checkout code
- uses: actions/checkout@v4

Trigger Types

1. push Runs the workflow when commits are pushed to the repository.

on:

push:

branches:

- main # Only runs on pushes to main
- 'release/*' # Runs on any branch starting with release/

paths:

- 'src/**' # Only runs if files in src/ are changed

- 'pom.xml' # Or if pom.xml changes

- branches — filters by branch name; supports glob patterns
- paths — filters by file path; workflow only runs if the changed files match
- tags — can also trigger on tag pushes: tags: ['v*']

2. pull_request Runs the workflow when a pull request is opened, updated, or closed.

on:

pull_request:

branches:

- main # Only for PRs targeting main

types:

- opened # When PR is first opened
- synchronize # When new commits are pushed to the PR branch
- reopened # When a closed PR is reopened

This is the most important trigger for CI — you want to run tests on every pull request before merging to ensure the code is safe to merge.

3. schedule Runs the workflow at a scheduled time using cron syntax.

on:

schedule:

- cron: '0 2 * * *' # Every day at 2:00 AM UTC
- cron: '0 9 * * 1' # Every Monday at 9:00 AM UTC

Cron format: minute hour day-of-month month day-of-week

Useful for:

- Nightly builds that run a full test suite
- Scheduled dependency vulnerability scans
- Periodic cleanup jobs

4. workflow_dispatch (Manual Trigger) Allows the workflow to be triggered manually from the GitHub UI or via the GitHub API.

on:

```
workflow_dispatch:
  inputs:
    environment:
      description: 'Target environment'
      required: true
      default: 'staging'
      type: choice
    options:
      - staging
      - production
  run_tests:
    description: 'Run tests before deploy?'
    required: false
    default: true
    type: boolean
```

When triggered manually, GitHub shows a form with the defined inputs. The values are accessible in the workflow as `${{ github.event.inputs.environment }}`.

5. `workflow_call` Allows one workflow to be called (reused) by another workflow — making it a reusable workflow.

on:

```
workflow_call:
  inputs:
    version:
      required: true
      type: string
```

6. `release` Runs when a GitHub release is created, edited, or published.

on:

```
release:
  types:
```

- published # When a release is published

Commonly used to trigger deployment pipelines when a new version is officially released.

7. Combining Multiple Triggers A single workflow can have multiple triggers:

on:

push:

branches: [main]

pull_request:

branches: [main]

schedule:

- cron: '0 0 * * 0' # Weekly on Sunday midnight

workflow_dispatch:
